

Five Days at Fremont — Part C

AC101 integrated Tesla case · worked in Lecture 3

*Part C of the week-long AC101 Tesla case — worked in Lecture 3. The full fact-vs-fiction note and company background are in the Part A handout (the complete case is on the course site). In short: Tesla is real and the filed figures are real; **every product-costing scenario, internal figure and decision here is hypothetical**; exhibits are labelled REAL, SIMULATED or ESTIMATE. Every question is worked and discussed in the lecture.*

Part C — Wednesday: one factory, five models, one pot of overhead

Wednesday morning, the question sharpens from “a Tesla” to “which Tesla”. The same factory floor you walked on Monday builds several models side by side — and someone in the pricing team has mailed around the per-car arithmetic with the subject line “why do we bother with the cheap ones?”

Stylised liberty, flagged: for a clean single-factory example, this part places all five models — Y, 3, S, X and Cybertruck — at Fremont. In reality the Cybertruck is built at Giga Texas, and the volumes here (1.23m cars a year) are stylised towards Tesla’s global model mix — about double the real plant’s ~650k installed annual capacity (per the 10-K). All internal figures in this part are simulated (Exhibits C-1 to C-3), and the margin levels are stylised at roughly 2.4× Tesla’s real 17.9% FY2024 gross margin — read the distribution across models, not the level.

The pricing team’s point looks unanswerable. Before any shared costs, a Model X clears \$67,000 per car over its direct materials (\$99,000 – \$32,000); a Model Y clears \$26,000 (\$44,000 – \$18,000). Pour the factory into premium cars, surely.

Except that between “price minus materials” and “profit” sits a **\$9,200m pot of shared overhead** — assembly lines, the battery plant, test labs, logistics, the building itself (Exhibit C-2). None of it can be traced to a single car; all of it must be *allocated* by a rule. And here is the uncomfortable discovery of Day 3: **whichever splitting rule you pick decides which model looks profitable.**

The simplest rule — and most firms’ default — is the **traditional volume allocation**: one plant-wide rate per car, on the logic “more cars, more cost”. Divide \$9,200m by 1,230,000 cars and every model, from a Model 3 to a Cybertruck, carries the same **\$7,480** of overhead. On that rule the premium cars look spectacular: margins around 60%, against roughly 42% for the volume cars (Exhibit C-3, top rows).

But walk the floor before you believe the spreadsheet. A Cybertruck spends 25 hours in assembly against a Model 3’s 10 — two and a half times as long. It carries a 123 kWh battery against 60. It goes through six test cycles against three. Does it *sip the same overhead* as a Model 3? Charging every car the same \$7,480 quietly makes the simple, high-volume cars pay for the complex, low-volume ones. That has a name — **cross-subsidisation** — and a cure.

Activity-based costing (ABC) splits the overhead into pools and charges each pool by the **driver** that actually causes it: assembly by assembly hours, the battery plant by kWh, testing by test cycles (Exhibit C-2 gives each pool and its driver total). Divide each pool by its driver total and you get a rate; charge each model for the drivers it consumes. The costing team ran exactly this before being pulled onto another project, leaving you Exhibit C-3 — complete except for the Cybertruck column, which is yours (Questions 2 and 3). The logic behind the drivers is Cooper and Kaplan’s **cost hierarchy**: some overhead is caused by each *unit* (assembly hours, kWh), some by each *batch*, some by a *product line* existing at all, and some by the *facility* existing — and a volume rate wrongly treats all four levels as unit-level, which is precisely why complex low-volume products come out under-costed.

When the column is filled in, the reveal is not that anything makes a loss – nothing does – but where the money actually lives. Per car, the premium models still earn the most. In *total*, they are a garnish: **Model Y and Model 3 together carry about 86% of the annual profit** in this stylised Fremont world (directionally right for the real Tesla, where the 3/Y pair is ~95% of volume). “Most profitable per car” and “where the money is” are different questions, and ABC sharpens the difference – deflating the premium margins and exposing the Cybertruck as earning far less than its sticker suggests.

Late on Wednesday, Reyes reads the table and asks the natural next question: “*If the Cybertruck’s margin is the weakest, why not drop it?*” Treat that question with respect – and suspicion. ABC allocates the Cybertruck roughly \$485m of overhead a year, but **allocated is not avoidable**: most of the \$9,200m is shared, committed capacity – the building, the lines, the battery plant – that stays whether or not the Cybertruck exists. Kill the model and, in the worst case (none of its overhead avoidable), the full \$485m reloads onto the survivors at about +\$407 per car – at which point another model “looks weak”, and the road ends in the **death spiral**: dropping products one by one because a spreadsheet rule keeps re-spreading the same fixed pot. Which costs would *actually* disappear is a different question from anything you have computed today. It is Thursday’s question.

Assignment questions – Part C

1. Compute the plant-wide blended rate from Exhibits C-1 and C-2. Under it, show the profit per car for the Model 3 and the Model X (price – materials – overhead). On these numbers, should Tesla pour the factory into premium cars? Hold your answer until Question 4.
2. Compute two ABC rates by calculator, exactly as the exam would ask: the **battery systems** rate per kWh and the **assembly** rate per hour. (The other three rates are given in Exhibit C-3.)
3. Fill in the Cybertruck column of Exhibit C-3: its ABC overhead per car, profit per car and margin. Then, in one paragraph, use **cross-subsidisation** and the **cost hierarchy** (unit/batch/product/facility) to explain why the volume rate flattered it.
4. Using C-3’s annual columns: where does the profit actually come from, and how do you reconcile “a Model X earns \$56,749 a car” with “Model Y and Model 3 are ~86% of the total”? Then the trap: the CFO suggests dropping the Cybertruck on its weak ABC margin. Its allocated overhead is ~\$485m a year – explain, pool by pool, why allocated ≠ avoidable, what the worst case does to the surviving models, and where that road ends.
5. ► **Discovery.** (*Hypothetical single-plant quarter, labelled illustrative: price \$45,000, variable cost \$30,000 per car, fixed production overhead \$1,200m, sales fixed by demand at 120,000 cars.*) Your bonus rides on reported (absorption-costing) profit, and you are about to miss. Show that producing 150,000 cars instead of 120,000 lifts reported profit from \$600m to \$840m while variable-costing profit stays at \$600m; trace exactly where the missing \$240m went. Name the practice, explain why it is legal yet value-destroying, and connect it to Roychowdhury (2006).
6. Before this morning’s allocation, the \$9,200m overhead pot had to be assembled from a longer working list. Exhibit C-4 shows the eight lines a junior analyst started from; the pot actually used in Questions 1 to 3 was \$9,200m. Three of these lines do not belong in a Fremont product-overhead pot. Identify the three, give a one-line reason for each, and confirm that the remaining lines reconcile to \$9,200m. Does removing them change the plant-wide rate of \$7,480 you found in Question 1?
7. The CFO drops the Cybertruck anyway, on its weakest ABC margin. Take the worst case from Question 4 – that none of its overhead is avoidable – so its full ~\$485m reloads flat across the 1,190,000 surviving cars, at ~+\$407 per car. Add that \$407 to each surviving model’s ABC overhead from Exhibit C-3, recompute the four margins, and identify which model the same “drop the weakest margin” rule now points at. State the one assumption your answer rests on, and explain in two sentences why running the rule a second time is self-destructive.

Exhibits – Part C

Exhibit C-1 – Fremont product data (stylised) SIMULATED – volumes, prices, materials and consumption figures built from industry benchmarks and teardown reports. Stylised liberty flagged: all five models are placed at Fremont for a clean single-factory example; in reality the Cybertruck is built at Giga Texas, and the 1.23m total is roughly double the real plant’s ~650k installed annual capacity (it leans towards Tesla’s global model mix).

Model	Annual volume	Price	Direct materials	Assembly hrs/car	Battery kWh	Test cycles
Model Y	650,000	\$44,000	\$18,000	12	75	3
Model 3	500,000	\$39,000	\$15,000	10	60	3
Model S	20,000	\$89,000	\$28,000	18	100	5
Model X	20,000	\$99,000	\$32,000	20	100	5
Cybertruck	40,000	\$80,000	\$35,000	25	123	6
Total	1,230,000					

Exhibit C-2 – Overhead cost pools and driver totals *SIMULATED* – pool sizes based on auto-industry benchmark cost shares. Note: logistics and facilities are charged per vehicle as proxies – logistics is really a batch-level cost (cars ship in multi-car transporter loads) and facilities a facility-level cost spread by volume, not a true square-foot driver.

Cost pool	Annual cost (\$m)	Cost driver	Driver total
Assembly	2,400	Assembly hours	14.56m hours
Battery systems	3,200	Battery kWh	87.67m kWh
Quality testing	800	Test cycles	3.89m cycles
Logistics	1,200	Vehicles	1.23m vehicles
Facilities	1,600	Vehicles	1.23m vehicles
Total	9,200		

Exhibit C-3 – The costing team's part-finished comparison *SIMULATED* – same stylised levels as C-1/C-2. Note carried from the costing team: margins here run ~2.4× Tesla's real 17.9% FY2024 gross margin – read the distribution across models, not the level. Three ABC rates are given: quality testing \$205.66 per cycle, logistics \$975.61 per vehicle, facilities \$1,300.81 per vehicle; the assembly and battery rates are Question 2, and the Cybertruck column is Question 3.

	Model Y	Model 3	Model S	Model X	Cybertruck
Traditional overhead per car	\$7,480	\$7,480	\$7,480	\$7,480	\$7,480
ABC overhead per car	\$7,609	\$6,732	\$9,922	\$10,251	[?]
Margin – traditional	42%	42%	60%	60%	47%
Margin – ABC	42%	44%	57%	57%	[?]
Annual ABC profit (\$bn)	12.0	8.6	1.0	1.1	[?]
Share of total ABC profit	50%	36%	4%	5%	[?]

Total annual ABC profit across all five models ≈ \$24.1bn. "Margin" = (price – materials – overhead) ÷ price.

Exhibit C-4 – The analyst's working list (before the pot was fixed) *SIMULATED* – illustrative Fremont-scale figures, not Tesla actuals.

Candidate line	\$m
Assembly	2,400
Battery systems	3,200
Quality testing	800
Logistics	1,200
Facilities	1,600
Head-office administration and national advertising	1,900
Paint-shop retooling – a one-off project completed last year	600
Battery-plant electricity (metered separately; forms part of the Battery systems pool above)	300
Working-list total	12,000